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1 The Local Theory of Curves

Definition 1.1 A curve $\vec{r}(t)\colon (a,b)\to\mathbb{R}^n$, where $\vec{r}(t)=(x_1(t),\dots,x_n(t))$, is *regular* if

(1) $x_i(t)\in C^\infty(a,b)$;

(2) $\forall t\in(a,b)$, $|\vec{r}\,'(t)|\neq 0$.

Specifically, we often take $n=2,3$.

Notations. $\vec{r}\,'=\frac{\mathrm{d}}{\mathrm{d}t}\vec{r}(t)$ and $\vec{r}\,'(s)=\frac{\mathrm{d}}{\mathrm{d}s}\vec{r}(s)$ for a regular curve $\vec{r}$, where s is called the arc length of $\vec{r}$, defined by the following formula:

$$s=\int_a^t |\vec{r}\,'(u)|\,\mathrm{d}u.$$

Hence

$$\frac{\mathrm{d}s}{\mathrm{d}t}=|\vec{r}\,'(t)|,$$

and

$$\vec{r}\,'(s)=\frac{\mathrm{d}\vec{r}}{\mathrm{d}s}=\frac{\mathrm{d}\vec{r}}{\mathrm{d}t}\cdot\frac{\mathrm{d}t}{\mathrm{d}s}=\vec{r}\,'\cdot\frac{1}{|\vec{r}\,'|}\quad\Longrightarrow\quad\forall s,\ |\vec{r}\,'(s)|=1.$$

Notations. Let $T(s)=\vec{r}\,'(s)$ be the tangent vector and $N(s)$ the normal vector with $|N|=1$. We assume that T is rotated counterclockwise by $\pi/2$ to obtain N.

1.1 Plane Curves

Let $\vec{r}=\vec{r}(s)$, $T(s)=\vec{r}\,'(s)$, $|T(s)|=1$. Then

$$|T(s)|^2=1\implies\frac{\mathrm{d}}{\mathrm{d}s}|T(s)|^2=0,\ \text{i.e.}\ \frac{\mathrm{d}}{\mathrm{d}s}\langle T,T\rangle=0\implies 2\langle T,T'\rangle=0\implies\langle T,T'\rangle=0,$$

i.e. $T'(s)=K_r(s)N(s)$, where K_r is some function, called the relative curvature of $\vec{r}$.

Lemma 1.2

$$\frac{\mathrm{d}}{\mathrm{d}s}\begin{pmatrix}T\\N\end{pmatrix}=\begin{pmatrix}0&K_r\\-K_r&0\end{pmatrix}\begin{pmatrix}T\\N\end{pmatrix}.$$

Proof. $T'(s)=K_rN(s)$. Since $|N(s)|=1$, $0=\langle N',N\rangle$. From $\frac{\mathrm{d}}{\mathrm{d}s}\langle T,N\rangle=0$ we get $\langle T',N\rangle+\langle T,N'\rangle=0$, hence

$$K_r\langle N,N\rangle+\langle T,N'\rangle=0\implies\langle T,N'\rangle=-K_r,$$

i.e. $N'(s)=-K_rT(s)$. ∎

Example 1.3 Given $\vec{r}(t) = (x_0 + R\cos t,\ y_0 + R\sin t)$ for $t \in (0, \pi)$, $R \in \mathbb{R}_+$, $(x_0, y_0) \in \mathbb{R}^2$, find $K_r(s)$.

Hint:

$$\vec{r}'(t) = R(-\sin t, \cos t) \implies |\vec{r}'(t)| = R = \frac{\mathrm{d}s}{\mathrm{d}t} \implies s = Rt.$$

Hence

$$\vec{r}(s) = \Big(x_0 + R\cos\frac{s}{R},\ y_0 + R\sin\frac{s}{R}\Big),$$

$$T(s) = \vec{r}'(s) = \Big(-\sin\frac{s}{R}, \cos\frac{s}{R}\Big),$$

$$T'(s) = \frac{1}{R}\Big(-\cos\frac{s}{R}, -\sin\frac{s}{R}\Big), \qquad N(s) = \Big(-\cos\frac{s}{R}, -\sin\frac{s}{R}\Big),$$

$$\implies K_r(s) = \frac{1}{R}.$$

Example 1.4 For a parameterized curve $\vec{r}(t) = (x(t), y(t))$,

$$K_r(t) = \frac{x'y'' - x''y'}{(x'^2 + y'^2)^{3/2}}.$$

1.2 Space Curves

Let $\vec{r}(s) = (x_1(s), x_2(s), x_3(s))$, $s \in I$ be a regular curve.

Definition 1.5 The curvature of $\vec{r}(s)$ is $K(s) = |\vec{r}''(s)|$. $\rho(s) = \frac{1}{K(s)}$ is called the radius of curvature.

Since $|\vec{r}'(s)| = 1$, i.e. $\langle \vec{r}'(s), \vec{r}'(s)\rangle = 1$, we have $\langle \vec{r}''(s), \vec{r}'(s)\rangle = 0$ for all s. Assume $K(s) \neq 0$; then

$$N(s) = \frac{\vec{r}''(s)}{|\vec{r}''(s)|} = \frac{\vec{r}''(s)}{K(s)}, \qquad \vec{r}''(s) = K(s)N(s), \qquad \text{i.e.} \quad \dot{T} = KN.$$

Definition 1.6 $B(s) = T(s) \times N(s)$ is called the binormal vector of $\vec{r}(s)$. Hence we have an orthogonal frame along $\vec{r}(s)$, called the Frenet frame: $\{\vec{r}(s); T, N, B\}$.

Definition 1.7 The torsion of $\vec{r}(s)$ is $\tau(s) = \langle \dot{N}, B\rangle$.

Remark 1.8 $\langle \dot{N}, B\rangle = \langle \dot{N}, T \times N\rangle = -\langle \dot{N}, N \times T\rangle = -\langle N, T \times \dot{N}\rangle$. Since

$$\langle N, \dot{B}\rangle = \langle \dot{T} \times N, N\rangle + \langle T \times \dot{N}, N\rangle = \langle KN \times N, N\rangle - \langle \dot{N}, B\rangle = -\langle \dot{N}, B\rangle,$$

we have $\tau(s) = \langle \dot{N}, B\rangle = -\langle N, \dot{B}\rangle$.

Theorem 1.9 (Serret–Frenet formulae) Let $\vec{r}(s)$, $s \in I$ be a regular space curve. Then

$$\frac{\mathrm{d}}{\mathrm{d}s}\begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & K & 0 \\ -K & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix}\begin{pmatrix} T \\ N \\ B \end{pmatrix}.$$

Proof. We have $\langle \dot{T}, T\rangle = \langle \dot{N}, N\rangle = \langle \dot{B}, B\rangle = 0$ and $\dot{T} = KN$. Since $\langle T, N\rangle = 0$,

$$\langle \dot{T}, N\rangle + \langle T, \dot{N}\rangle = 0 \implies \langle \dot{N}, T\rangle = -K.$$

And $0 = \langle B, T\rangle$ gives

$$0 = \langle \dot{B}, T\rangle + \langle B, \dot{T}\rangle = \langle \dot{B}, T\rangle + \langle B, KN\rangle,$$

hence $\langle \dot{B}, T\rangle = 0$.

Finally, $\dot{N} \perp N$ and $\dot{B} \perp B$ since $|N| = |B| = 1$, so

$$\dot{N} = \langle \dot{N}, T\rangle T + \langle \dot{N}, B\rangle B = -KT + \tau B, \qquad \dot{B} = \langle \dot{B}, T\rangle T + \langle \dot{B}, N\rangle N = -\tau N,$$

where $\langle \dot{N}, B\rangle = \tau$ by definition and $\langle \dot{B}, N\rangle = -\tau$ by Remark 1.8. ∎

Remark 1.10 Note that $\dot{T}(s) = \vec{r}''(s) = K(s)N(s)$ and $\dot{N}(s) = -K(s)T(s) + \tau(s)B(s)$, so

$$\vec{r}'''(s) = \dot{K}N + K\dot{N} = \dot{K}N - K^2T + K\tau B.$$

It is worth noting that

$$\dot{N} = \frac{1}{K}(\vec{r}''' - \dot{K}N).$$

Hence

$$\tau = -\langle \dot{B}, N\rangle = \langle B, \dot{N}\rangle = \frac{1}{K}(\langle B, \vec{r}'''\rangle - \dot{K}\langle B, N\rangle) = \frac{1}{K}\langle B, \vec{r}'''\rangle.$$

That is,

$$\tau(s) = \frac{1}{K^2(s)}\langle \vec{r}' \times \vec{r}'',\ \vec{r}'''\rangle.$$

Example 1.11 The circular helix: $\vec{r}(t) = (R\cos t,\ R\sin t,\ bt)$, $R \in \mathbb{R}_+$, $b \in \mathbb{R}$.

Proposition 1.12 A space curve is a plane curve iff $\tau(s) = 0$ for all $s \in I$.

Proof. 1° "$\Longrightarrow$". If $\vec{r}(s)$, $s \in I$ is a plane curve, then there are $\vec{n}_0 \in \mathbb{R}^3$, $|\vec{n}_0| = 1$ and $x_0 \in \mathbb{R}^3$ such that $\langle \vec{n}_0, \vec{r}(s) - x_0\rangle = 0$. Hence

$$\langle \vec{n}_0, T(s)\rangle = 0 \implies \langle \vec{n}_0, \vec{r}''(s)\rangle = 0,$$

i.e. $\langle \vec{n}_0, N(s)\rangle = 0$. Thus $\vec{n}_0 \perp T$, $\vec{n}_0 \perp N$, so $\vec{n}_0 \parallel B$, and $\tau = -\langle \dot{B}, N\rangle = 0$.

2° "$\Longleftarrow$". If $\tau(s) = 0$ for all $s \in I$, then $0 = \tau(s) = -\langle \dot{B}, N\rangle$. Since $0 = \langle \dot{B}, B\rangle$, we have $\dot{B} \perp B$; write $\dot{B} = \alpha T$. As

$$\langle \dot{B}, T\rangle = -\langle B, \dot{T}\rangle = -\langle B, KN\rangle = 0,$$

we get $\alpha = 0$, hence $\dot{B}(s) = 0$. Say $B(s) = \vec{n}_0$ for some $\vec{n}_0$, $|\vec{n}_0| = 1$. By definition

$$0 = \langle B(s), T(s)\rangle = \langle \vec{n}_0, \vec{r}\,'(s)\rangle = \frac{\mathrm{d}}{\mathrm{d}s}\langle \vec{n}_0, \vec{r}(s) - x_0\rangle,$$

so $\vec{r}(s)$, $s \in I$ is a plane curve. ■

Definition 1.13 A curve is called a *helix* if the tangent lines of $\vec{r}(s)$ make a constant angle with a fixed direction.

Lemma 1.14 Assume $\tau(s) \neq 0$ for all $s \in I$. Then $\vec{r}$ is a helix iff $\frac{K(s)}{\tau(s)} = \text{const}$ for all $s \in I$.

Proof. 1° "$\Longrightarrow$". If $\vec{r}$ is a helix, then there is $\vec{a} \in \mathbb{R}^3$, $|\vec{a}| = 1$, such that $\langle T(s), \vec{a}\rangle = \text{const}$, hence $\langle \dot{T}, \vec{a}\rangle = 0$, i.e. $\langle N(s), \vec{a}\rangle = 0$ for all s.

That is to say, $\vec{a} = \cos\theta\, T + \sin\theta\, B$ for some $\theta = \theta(s)$. Then

$$0 = -\sin\theta\,\dot{\theta}\,T + \cos\theta\,\dot{T} + \cos\theta\,\dot{\theta}\,B + \sin\theta\,\dot{B} = \dot{\theta}(-\sin\theta\,T + \cos\theta\,B) + \cos\theta\,(KN) + \sin\theta\,(-\tau N).$$

Hence $\dot{\theta} = 0$, so $\theta = \theta_0$ for some constant θ_0, and $K\cos\theta - \tau\sin\theta = 0$, i.e. $\tan\theta = \frac{K}{\tau}$. Thus

$$\frac{K}{\tau} = \tan\theta_0 = \text{const}.$$

2° "$\Longleftarrow$". By assumption $\frac{K}{\tau} = \text{const} = \tan\theta$ with θ fixed. Take $\vec{a}(s) = \cos\theta\, T + \sin\theta\, B$, as in 1°. Then $\vec{a}(s) \equiv \vec{a}_0$, so $\vec{r}(s)$ is a helix. ■

Theorem 1.15 (Fundamental theorem for the local theory of curves) Given smooth function $K(s)$, $\tau(s)$ on (a, b) with $K(s) > 0$, then there exists $\vec{r}(s)$ has arc length s with curvature $K(s)$ and torsion $\tau(s)$. Moreover, such a curve is unique up to a rigid motion of $\mathbb{R}^3$: for any $s_0 \in (a, b)$ and any positively oriented orthonormal frame $\{x_0; T_0, N_0, B_0\}$, there is exactly one such curve with

$$\vec{r}(s_0) = x_0, \quad T(s_0) = T_0, \quad N(s_0) = N_0, \quad B(s_0) = B_0.$$

Exercise 1.16 For $\vec{r}(s) \in \mathbb{R}^3$ be a curve, $s \in I$, we have

$$K(s) = \frac{|\vec{r}\,' \times \vec{r}\,''|}{|\vec{r}\,'|^3}, \qquad \tau(s) = \frac{\langle \vec{r}\,' \times \vec{r}\,'',\ \vec{r}\,'''\rangle}{|\vec{r}\,' \times \vec{r}\,''|^2}.$$